

CSCI 5402

Research Methods in Human-Robot Interaction

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Applied Scientist (70.6 / 100)	59.4 / 100

59	92	+43
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

Introduces students to the field of human-robot interaction (HRI). Covers HRI theory, principles, methodologies, and applications with links to robotics, artificial intelligence, human factors, human-computer interaction, design, cognitive psychology, education and other domains. Coursework includes readings from state-of-the-art in HRI research, team exercises and problem-solving sessions, and implementation and evaluation of a human-robot interaction systems for specific applications.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Applied Scientist	70.6	100.0	+29.4	MEDIUM
AI Engineer	68.6	94.1	+25.5	MEDIUM
Machine Learning Engineer	59.7	90.7	+31.0	HIGH
ML Engineer	59.6	88.1	+28.5	MEDIUM
Data Scientist	51.1	93.9	+42.8	HIGH
MLOps Engineer	46.6	85.5	+38.8	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Business Communication 2 of 6 roles

insights business

Deep Learning 2 of 6 roles

cnn training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **MEDIUM** **Consider Business Communication content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.

2 **MEDIUM** **Consider Deep Learning content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: cnn, training.

GAP KEYWORDS BY ROLE

Applied Scientist

70.6 → 100.0 (+29.4 pts)

discovery scientific reasoning literature problems models technical data

AI Engineer

68.6 → 94.1 (+25.5 pts)

perception data models learning infrastructure form software technical

Machine Learning Engineer

59.7 → 90.7 (+31.0 pts)

data interfaces sets learning analytics tools exploratory business

ML Engineer

59.6 → 88.1 (+28.5 pts)

cnn discovery data learning machine models infrastructure problems

Data Scientist

51.1 → 93.9 (+42.8 pts)

data sets analytics insights social problems approaches

MLOps Engineer

46.6 → 85.5 (+38.8 pts)

infrastructure detection data text learning machine image training

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.